Adaptive Restart and Nogood Management for Constraint Satisfaction Solvers

A Quantitative Risk & Finance-Inspired Search Control Architecture in Glucose 4.2.1

DAC - 203: ARTIFICIAL INTELLIGENCE

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Research Overview & Quantitative Architecture Dashboard

High-level problem, quantitative methodology, and rigorously verified empirical findings

❑ **THE CORE CHALLENGE**
❑ **Search Stagnation:**
❑ Classical CDCL solvers suffer from heavy-tailed runtimes when trapped in futile subtrees.
❑ **Rigid Static Policies:**
❑ Luby restart sequences operate blind to real-time search volatility.
❑ **Core Question:**
❑ **Can financial risk metrics dynamically optimize search control?**

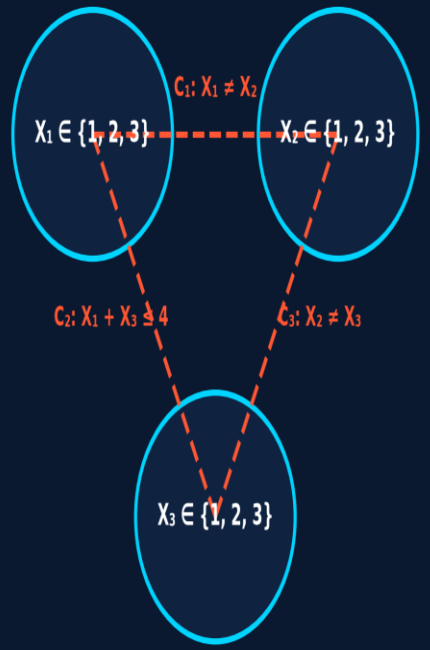
❑ **QUANTITATIVE METHODS**
❑ **V3 Drawdown Model:**
❑ Tracks conflict return rt via dual EWMAs (fast=0.08, slow=0.01) and decaying HWM.
❑ **Adaptive-Fixed Retention:**
❑ Continuous scoring: R(c) = 0.50 LBD + 0.40 Act + 0.10 Size with decoupled Glucose cadence.
❑ **Quantitative Risk Engine:**
❑ Momentum (Sharpe < -0.35) and tail shock (VaR95) with 1500-conflict gating.
❑ **Portfolio-MAB:**
❑ Two-armed bandit for online restart policy switching.

❑ **VERIFIED RESULTS**
❑ **Development Suite Winner:**
❑ **V3 Drawdown + AdaptFix won the 36-benchmark development suite (PAR-2 2.5381s vs 2.7463s baseline; -15.5% conflicts).**
❑ **Holdout Suite Winner:**
❑ **Glucose + AdaptFix won the 48-benchmark holdout suite (PAR-2 0.1107s; 144/144 solved).**
❑ **Hanoi-5 Planning Speedup:**
❑ V3 achieved 80.0% speedup over No-Heuristic control with surgical 5 restarts (3.06s vs 15.28s).

Constraint Satisfaction & Boolean Satisfiability

Combinatorial complexity, CNF representation, and industrial relevance

General Constraint Satisfaction (CSP)



Arbitrary finite or continuous domains
Multi-arity non-linear constraints

Boolean Satisfiability (CNF SAT)

CNF Formula: ∧ Clause j

$$(X_1 \vee \neg X_2 \vee X_3)$$

$$\wedge\ (\neg X_1 \vee X_4)$$

$$\wedge\ (X_2 \vee \neg X_3 \vee \neg X_4)$$

Variables: x_i ∈ {False (0), True (1)}

Canonical Boolean CSP Representation
Enables fast 2-Watched Literal & 1-UIP CDCL

❑ GENERAL CONSTRAINT SATISFACTION (CSP)

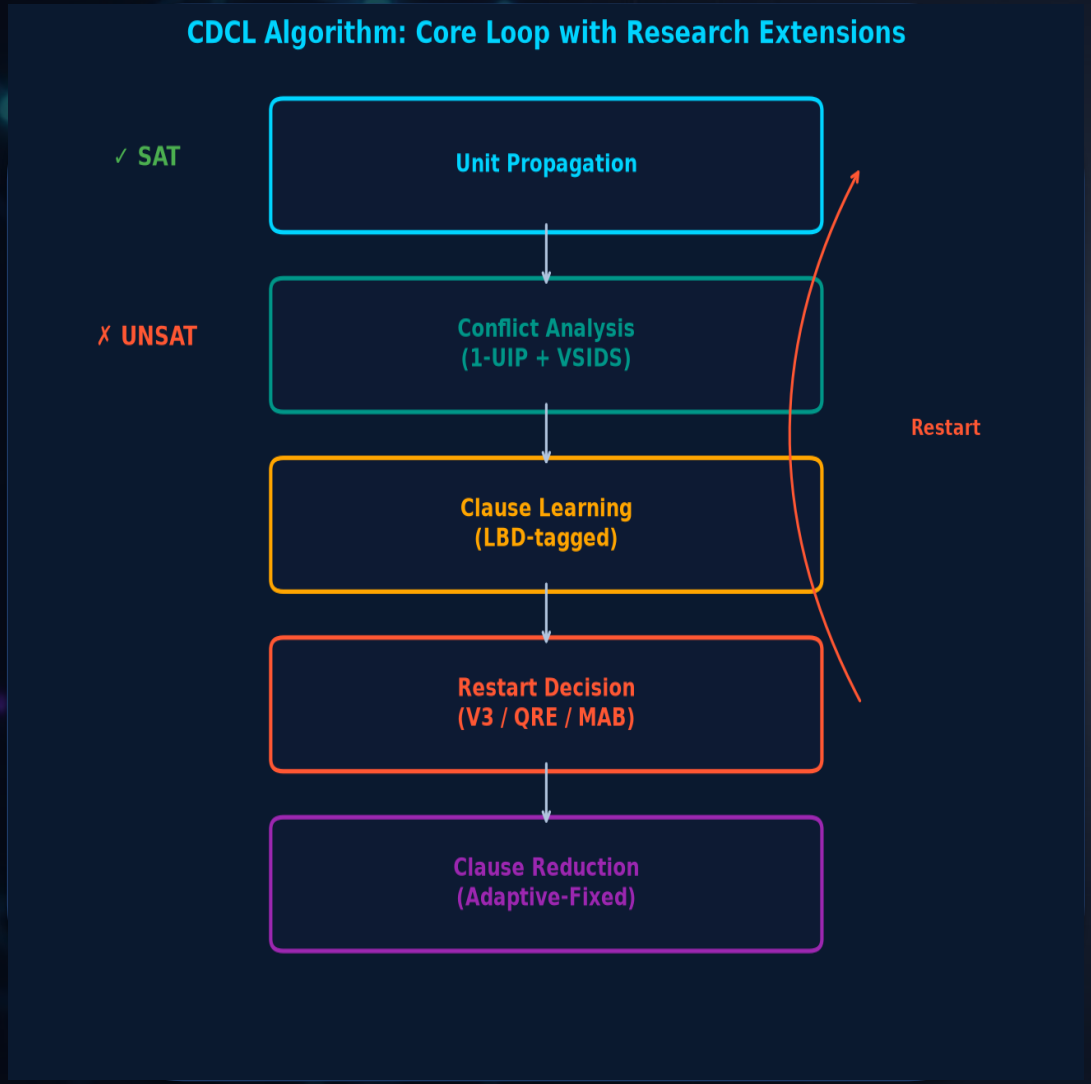
- Formal Definition: Tuple <X, D, C> where X is variables, D is domain, C is constraints.
- Combinatorial Complexity: Finding a consistent assignment is NP-complete in general.
- **Ubiquitous Real-World Applications:**
- Resource allocation, job-shop scheduling, vehicle routing, frequency assignment, and circuit layout.

❑ BOOLEAN SAT: THE CANONICAL CORE

- Canonical Representation: Discrete CSPs encode into CNF SAT via Direct or Logarithmic encodings.
- Cook-Levin Theorem (1971): Established Boolean SAT as foundational NP-complete problem.
- **Industrial Significance:**
- Hardware formal verification (Intel, AMD), software model checking, cryptographic analysis, and automated AI planning.
- **The Scalability Challenge: Modern formulas contain millions of variables; naive backtracking fails completely.**

Inside Modern Conflict-Driven Clause Learning (CDCL)

Step-by-step execution cycle of modern industrial SAT solvers



01 **VSIDS Variable Branching**
Decays variable conflict activity by 0.95, focusing search on variables in active contention.

02 **Two-Watched Literals (2WL)**
Inspects clauses only when a watched literal is false, enabling tens of millions of propagations/s.

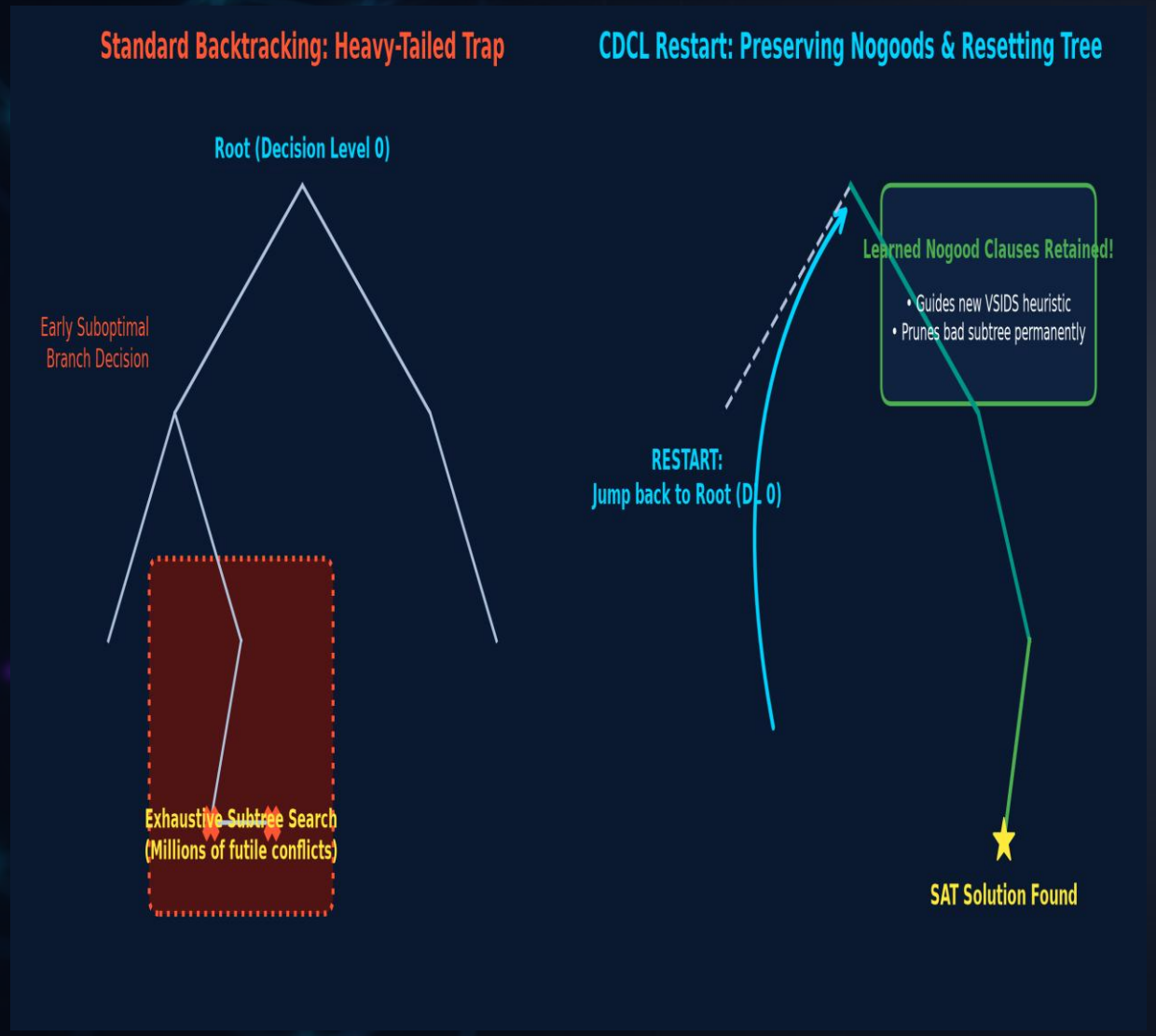
03 **1-UIP Conflict Analysis**
Traverse implication graph backwards to First Unique Implication Point to derive asserting nogood.

04 **Non-Chronological Backjumping**
Jumps directly to the asserting level of learned clause, permanently pruning unsatisfiable subtrees.

05 **Search Control: Restarts & Reduction**
Periodically cancels trail to escape local traps; regularly purges low-quality learned clauses.

The Dual Search-Control Dilemmas: Restarts & Clause Bloat

Why static restart heuristics and unchecked nogood accumulation degrade solving efficiency



❑ **DILEMMA I: THE RESTART DILEMMA**

➢ **Heavy-Tailed Traps (Gomes et al.):**

➢ Backtrack search runtimes follow heavy-tailed Pareto distributions. Early poor decisions trap solvers in massive futile subtrees.

➢ Premature Restart Risk: Erases partial assignment trails before deep implications are uncovered, causing search thrashing.

➢ Delayed Restart Risk: Static Luby sequences operate blind to real-time search state, wasting millions of conflicts in barren regions.

❑ **DILEMMA II: CLAUSE DATABASE BLOAT**

➢ **Unbounded Clause Growth:**

➢ Learning one clause per conflict rapidly generates hundreds of thousands of learned nogoods, exhausting working memory.

➢ **Propagation Latency Degradation:**

➢ **Excessive clauses bloat Two-Watched Literal lists, slowing BCP throughput by up to 300% on complex industrial formulas.**

➢ Information Loss Tradeoff: Deleting too aggressively purges critical lemmas, forcing repeated rediscovery of conflict subtrees.

Four Foundational Pillars in Combinatorial Search

Seminal papers informing our adaptive restart and clause retention framework

Audemard & Simon (IJCAI 2009 / JSAT 2012)

Glucose: Predicting Learned Clause Quality Using LBD

- **Introduced Literal Block Distance (LBD): Number of distinct decision levels in a clause.**
- Core Insight: Clauses with LBD <= 2 ('Glue Clauses') are exceptionally powerful and must never be deleted.
- Dynamic Restarts: Luby sequence with moving average LBD comparisons to trigger rapid clearing.

Biere & Fröhlich (SAT 2018)

CaDiCaL: Chronological Backtracking & Inprocessing

- **Demonstrated that non-chronological backjumping can be augmented with chronological steps.**
- Highlighted the vital role of interleaved in processing and bounded variable elimination.
- Emphasized disciplined learned-clause garbage collection cadences.

Liang, Cherif, & Li (CP 2016 / 2026)

MAB Learning at the Right Time Scale for SAT

- **Applied Multi-Armed Bandits to select variable branching heuristics (CHB / LRB).**
- Key Lesson: Reward signals must match the temporal scale of search decisions.
- Showed non-stationary bandit algorithms track solver search mode shifts.

Tolpin & Shimony (2011) / Koriche et al. (2022)

Value of Information (VOI) and Adaptive Tournaments

- **Formally modeled the computational cost-benefit tradeoff of heuristic evaluations.**
- Adaptive Tournaments (AST): Eliminates suboptimal search policies early via short runs.
- Demonstrated online meta-heuristics can outperform statically tuned solvers.

Research Objectives & Technical Hypotheses

Three core questions driving our quantitative search control architecture



❑ **Research Question 1 (Adaptive Restarting): Can finance-inspired drawdown metrics (dual EWMA, High-Water Mark) detect search stagnation better than static Luby sequences?**

➢ Hypothesis: Monitoring propagation returns and drawdown collapse (Dt >= 0.75 for 30 consecutive conflicts) will prune futile subtrees while preserving productive momentum.

❑ **Research Question 2 (Adaptive Clause Retention): Can composite multi-attribute clause scoring (LBD + activity + size) resolve database bloat without causing legacy reduction deferral bugs?**

➢ Hypothesis: A decoupled Glucose cadence (2000+300n) coupled with R(c) = 0.50 LBD + 0.40 Act + 0.10 Sz will maintain lean memory and cut total conflicts.

❑ **Research Question 3 (Online Policy Portfolio): Can a non-stationary Multi-Armed Bandit (Exp3 / D-UCB) adaptively switch between exploration and bounded descent during search?**

➢ Hypothesis: Online learning on conflict-normalized rewards will dynamically match the optimal restart policy to instance difficulty.

Reference Algorithms in Native Glucose 4.2.1

The reference components against which all research modifications are evaluated

1. VSIDS Variable Branching

- Variable State Independent Decaying Sum tracks conflict involvement.
- Decay Mechanism: Variable activity bumped upon conflict, multiplied by var_decay = 0.95.
- **Phase Saving: Reassigns variable to its previous polarity upon restart.**

2. Two-Watched Literals (2WL)

- Watches exactly 2 unassigned literals per clause in memory.
- Zero Overhead on False: Literals assigned false do not require list updates unless watched.
- **Enables tens of millions of propagations per second in C++.**

3. Glucose Dynamic LBD Restarts

- Computes Literal Block Distance (LBD) for every learned conflict clause.
- Moving Average Comparison: Fast LBD queue (size 50) vs slow cumulative average.
- **Trigger: If fast_avg * 0.80 > slow_avg, rapid restart is triggered to clear shallow trail.**

4. Baseline Clause Reduction (ReduceDB)

- Triggered at fixed intervals: firstReduceDB = 2000, incReduceDB = 300.
- Strict Protection: Clauses with LBD <= 2 (Glue clauses) are permanently preserved.
- **Sorting: Non-glue clauses sorted solely by VSIDS activity; bottom 50% purged.**

Adaptive Restart V1 & V2: Signal-Based Search Control

Initial implementations targeting propagation efficiency and anti-starvation ceilings

ADAPTIVE RESTART V1: PROGRESS RATIO

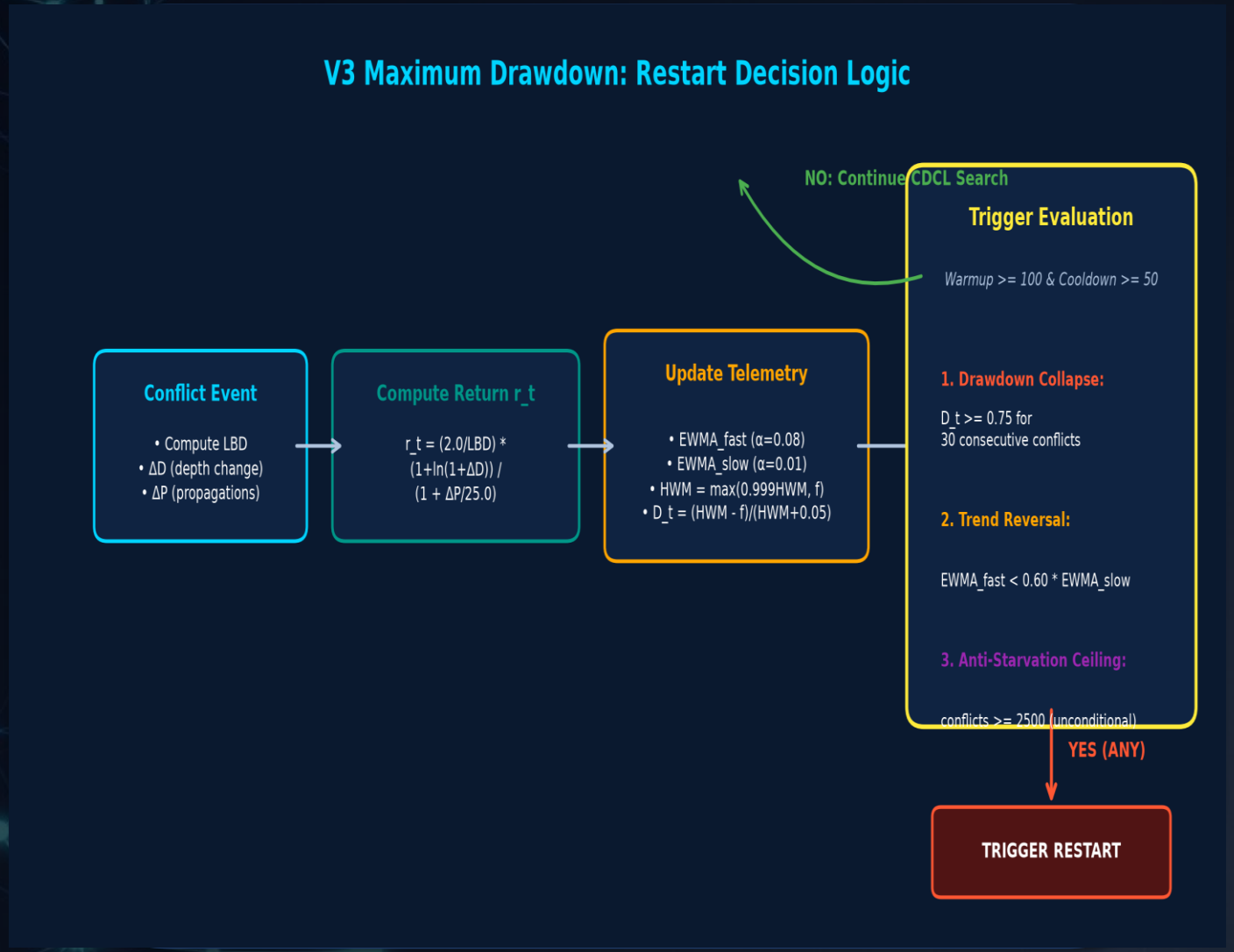
- **Core Efficiency Score (SW):**
- Monitors unit propagation yield per decision level in a rolling circular buffer (ADAPT_WIN_CAPACITY = 50 conflicts):
- **S_W = μ_{P,W} / (σ_{L,W} + 1.0)**
- where μ_{P,W} is mean propagations per conflict and σ_{L,W} is level standard deviation.
- **Global Baseline (S0):**
- Cumulative baseline tracked via Welford's online variance algorithm from search start.
- **Trigger Condition: S_W < 0.60 * S_0 (after 100 warmup and 50 min interval conflicts).**
- **V1 Limitation (Starvation Trap):**
- On certain structured formulas, SW remained permanently above threshold, allowing the solver to run for 50,000+ conflicts without restarting!

ADAPTIVE RESTART V2: ANTI-STARVATION

- **V2 Anti-Starvation Ceiling:**
- Forces an unconditional restart if search continues too long without progress:
- **adapt_conflicts_since_restart >= 2500**
- Guarantees the solver never remains trapped in a deep unsatisfiable subtree indefinitely.
- **V2 Quality Fallback Trigger:**
- Fallback check based on short-term LBD deterioration:
- **(lbdQueue.getavg() * 0.80) > (sumLBD / conflictsRestarts)**
- Requires 100 warmup and 50 cooldown conflicts.
- **Experimental Takeaway: V2 prevented starvation, but was purely reactive and lacked knowledge of historic peak performance.**

V3 Maximum Drawdown: Concept & Financial Mapping

Mapping portfolio drawdown theory to combinatorial search-space stagnation



Portfolio Asset Return (rt)
Quality-adjusted propagation yield and backjump depth per conflict.

Fast EWMA (Short-term)
Recent solver trajectory (α = 0.08, ~12 conflict memory).

Slow EWMA (Long-term)
Baseline solver health (α = 0.01, ~100 conflict memory).

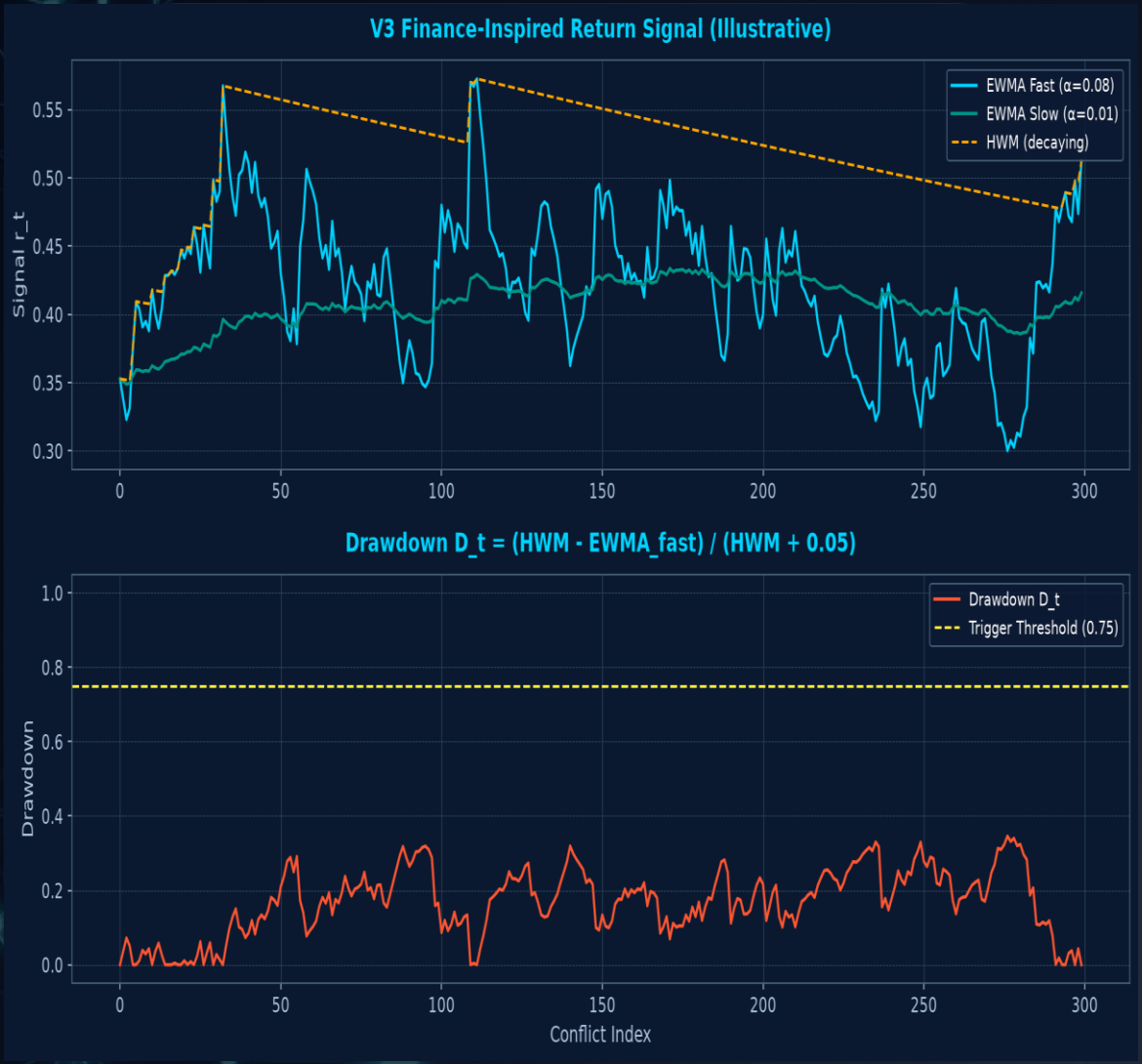
High-Water Mark (HWM)
Peak search efficiency achieved, decaying at 0.1%/step.

Drawdown (Dt)
Relative percentage drop from peak performance: (HWM - Fast) / (HWM + 0.05).

Stop-Loss (Restart)
Cut search losses when drawdown >= 0.75 persists for 30 conflicts.

V3 Maximum Drawdown: Exact Formulation & Parameters

Exact C++ implementation cross-verified against Solver.cc lines 3086-3258



EXACT VERIFIED FORMULAS (Solver.cc)

1 Quality-Adjusted Conflict Return (rt):

r_t = (2.0 / max(1, LBD)) * (1.0 + ln(1.0 + ΔD)) / (1.0 + ΔP / 25.0)

Verified: Uses logarithmic depth boost and linear propagation cost.

2 Dual-Horizon EWMAs:

EWMA_fast = 0.92 * EWMA_fast + 0.08 * r_t (α_fast = 0.08)

EWMA_slow = 0.99 * EWMA_slow + 0.01 * r_t (α_slow = 0.01)

3 Decaying High-Water Mark (HWM):

HWM_t = max(0.999 * HWM_{t-1}, EWMA_fast_t) (0.1% decay per conflict)

4 Relative Drawdown & Verified Trigger:

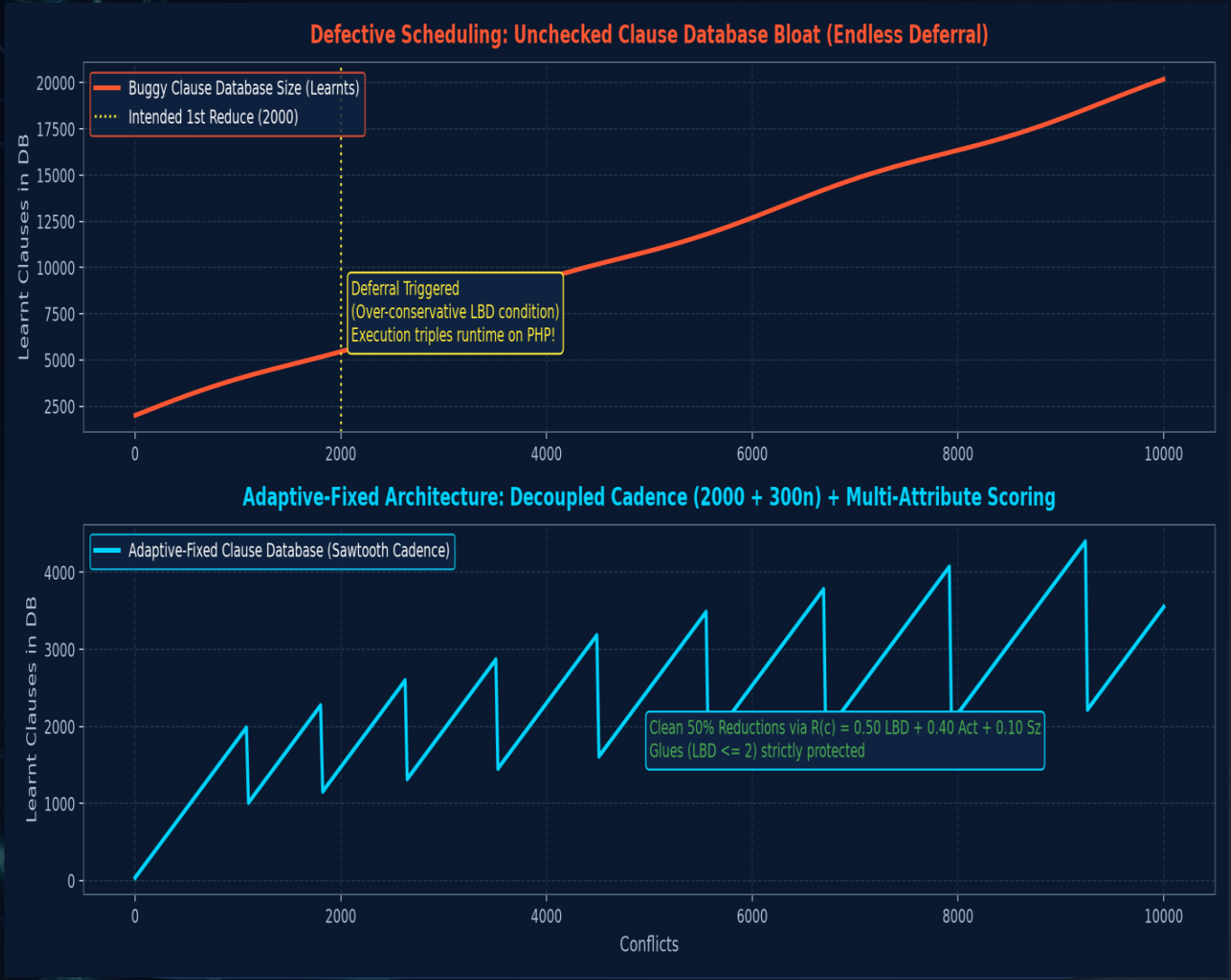
D_t = (HWM_t - EWMA_fast_t) / (HWM_t + 0.05)

Trigger: v3_consec_drawdown >= 30 with D_t >= 0.75 (requires 30 consecutive conflicts, min cooldown 50 conflicts, warmup 100).

5 Auxiliary Triggers: Trend reversal (fast < 0.60 * slow) | Ceiling (conflicts >= 2500).

Learned-Clause Management & The Deferral Defect

Discovery and resolution of the critical reduction scheduling deferral defect

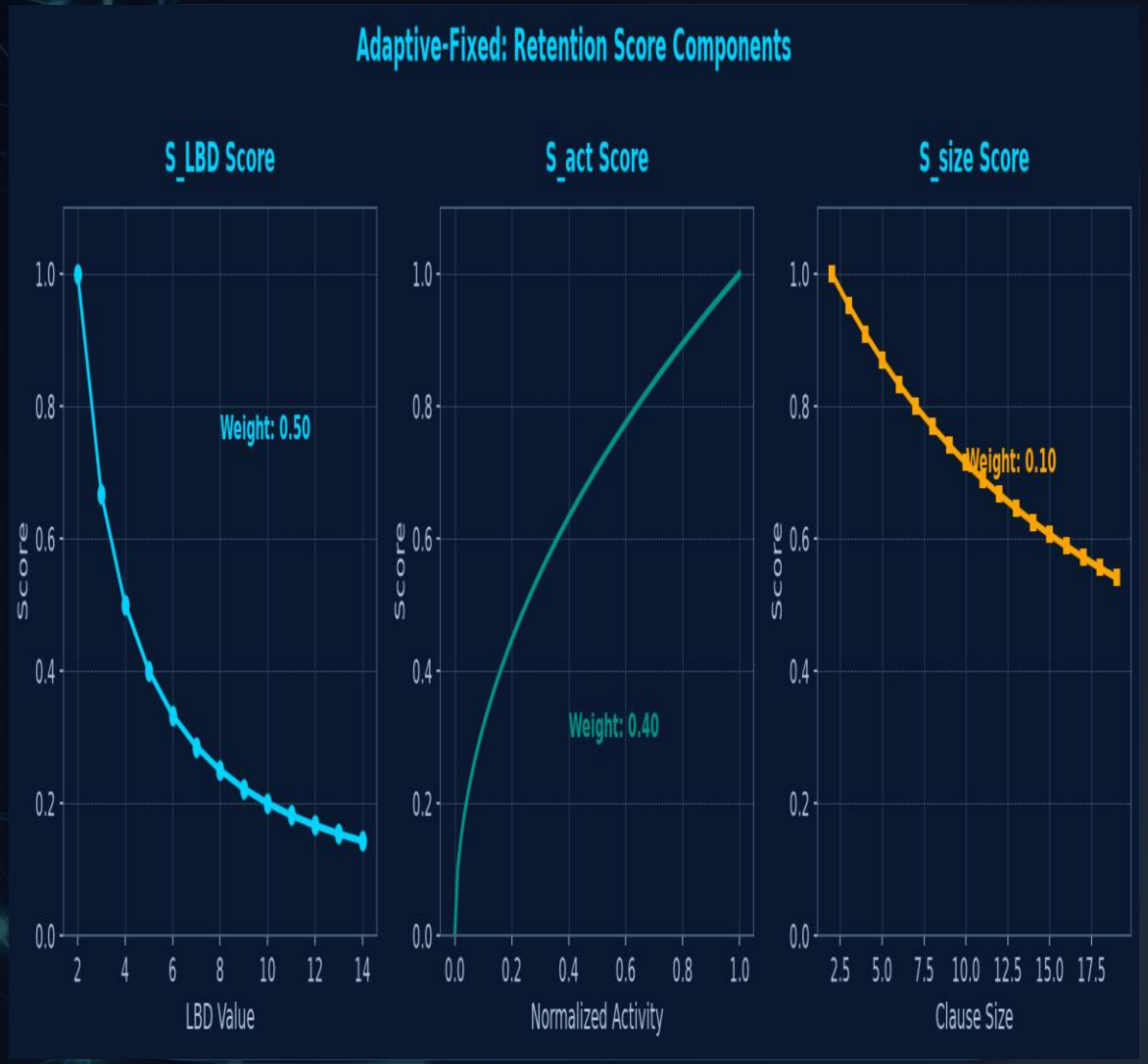


THE SCHEDULING DEFECT DISCOVERY

- **The Pathology Discovered:**
- In Step 32-34 experiments, legacy adaptive clause reduction tripled runtimes on Pigeonhole (PHP) formulas compared to baseline Glucose.
- **Root Cause Analysis:**
- The reduction trigger relied on an over-conservative LBD variance check that continuously postponed database cleanup:
- **nbclausesbeforereduce += 1000 was executed infinitely, bloating learned clauses past 100,000!**
- Severe Consequences: 2WL linked-list traversal overhead collapsed BCP throughput from 2.5M props/s to <400k props/s.
- **The Architectural Solution:**
- **Decoupled ReduceDB scheduling entirely from adaptive restart state, restoring Glucose's predictable 2000+300n cadence.**

Adaptive-Fixed Retention: Tri-Criteria Scoring

Continuous multi-attribute scoring function with strict protections (Solver.cc:1661-1825)

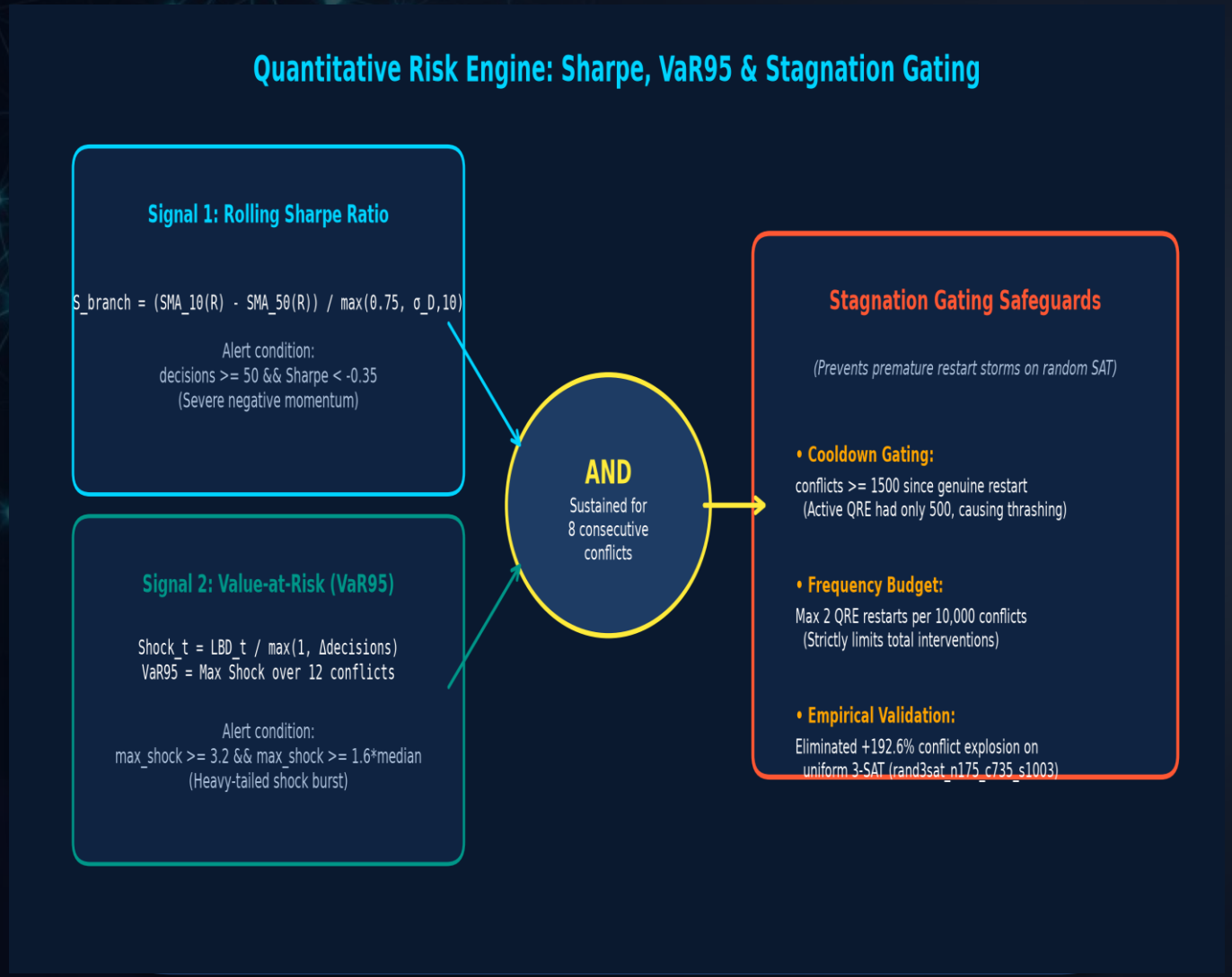


COMPOSITE SCORING R(c)

- **R(c) = 0.50 * S_LBD + 0.40 * S_act + 0.10 * S_size**
- **1 LBD Score (50% Weight):**
- S_LBD = 1.0 / (1.0 + 0.50 * max(0, LBD - 2))
- LBD=2 gives 1.0; LBD=4 gives 0.50; LBD=8 gives 0.25. Dominant quality signal.
- **2 Activity Score (40% Weight):**
- S_act = sqrt(clamp(activity / max_activity, 0.0, 1.0))
- Square-root mapping gives fair retention credit to recently active clauses.
- **3 Size Score (10% Weight):**
- S_size = 1.0 / (1.0 + 0.05 * max(0, size - 2))
- **Strict Protection Classes (Never Deleted):**
- Binary clauses (size <= 2) • Glue clauses (LBD <= 2)
- Single-round immune clauses (!canBeDel) • Reason clauses (locked)
- **Proven Result: Won Holdout Suite (PAR-2 0.1107s; 144/144 solved).**

Quantitative Risk Engine: Sharpe Momentum & Tail Shock

Rolling Sharpe ratio and non-parametric Value-at-Risk signals for emergency intervention



MATHEMATICAL TELEMETRY SIGNALS

- **Signal 1: Rolling Sharpe Ratio:**
- Measures search momentum vs depth volatility:
- **S_branch = (SMA_10(R) - SMA_50(R)) / max(0.75, σ_D,10)**
- Alert: decisions >= 50 && Sharpe < -0.35 indicates severe collapse in propagation yield.
- **Signal 2: Value-at-Risk (VaR95):**
- Tracks conflict shock: Shock_t = LBD_t / max(1, Δdecisions).
- **VaR95 = Max Shock over 12 conflicts**
- Verified: Non-parametric nearest-rank maximum shock over a 12-conflict window (not Gaussian).
- **Alert: max_shock >= 3.2 && max_shock >= 1.6 * median.**
- **Conjunction: Emergency restart requires BOTH signals active for 8 consecutive conflicts.**

Stagnation-Gated QRE: Taming Random SAT Cascades

Identifying hyperactive restart pathology on random SAT and enforcing strict gating

THE RANDOM SAT ANOMALY (ACTIVE QRE)

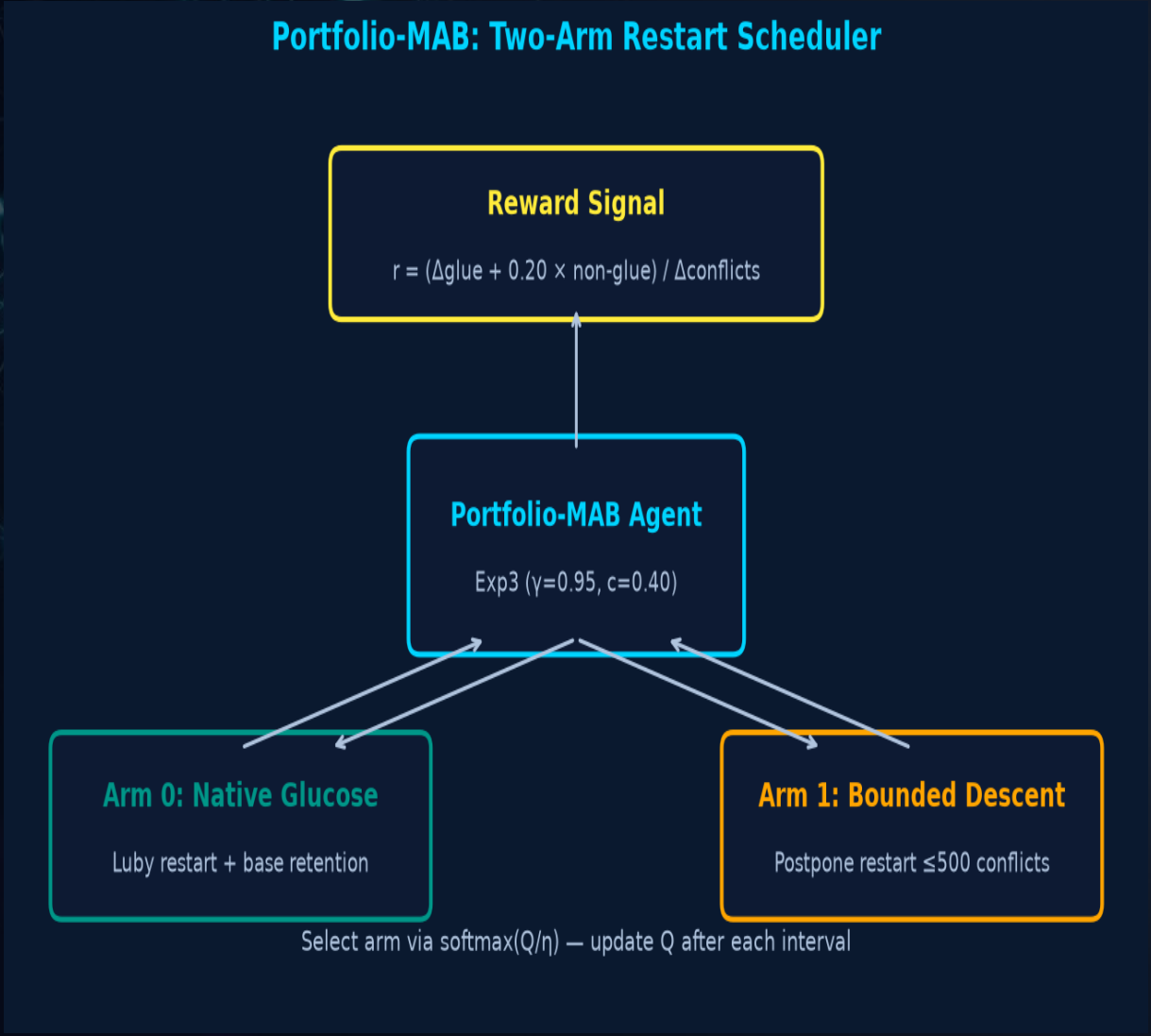
- **Hyperactive Restart Pathology:**
- On uniform random 3-SAT formulas near phase transition (rand3sat_n175_c735_s1003.cnf), Active QRE caused restart thrashing.
- **Severe Conflict Explosion:**
- **Total conflicts exploded by +192.6% compared to native Glucose baseline!**
- **Root Cause Analysis:**
- In uniform random SAT, variable activity is homogeneous. Local statistical variance triggered false VaR95 alerts, resetting the trail before VSIDS could concentrate on backdoor variables.
- Shadow Telemetry Overhead: Running QRE calculations in shadow mode added 14.9% CPU overhead with zero solver decisions altered.

STAGNATION-GATING STABILIZATION

- **Safeguard 1: Extended Cooldown:**
- Increased minimum conflicts between genuine restarts from 500 to 1,500 conflicts:
- **conflicts_since_genuine_restart >= 1500**
- **Safeguard 2: Frequency Budget:**
- **Enforced a strict mathematical ceiling of max 2 QRE restarts per 10,000 conflicts.**
- **Safeguard 3: Sustained Conjunction:**
- Both Sharpe and VaR must remain active for 8 consecutive conflicts.
- **Empirical Outcome: Stagnation-Gating completely suppressed restart thrashing on random SAT, reducing conflicts by 850k across the suite.**

Duration-Aware Portfolio-MAB: Two-Armed Bandit Scheduler

Online non-stationary algorithm selection using conflict-normalized rewards (Step 44)



EXP3 / D-UCB TWO-ARM ARCHITECTURE

- **Arm Definition:**
- Arm 0: Native Glucose dynamic LBD restarts.
- Arm 1: Bounded descent (postpones restart by up to Δc_max = 500 conflicts).
- **Conflict-Normalized Reward (Verified):**
- **Reward = clamp((Δglue + 0.20 * Δnon_glue) / Δc, 0.0, 1.0)**
- Normalized by conflicts elapsed Δc (not wall-clock time as initially drafted).
- Discounted Selection: γ = 0.95, c = 0.40.
- **Pilot Findings (16 instances, 240 runs): Promising PAR-2 (0.1243s vs 0.1390s Glucose). All 240 runs passed verification.**
- **Disciplined Academic STOP Decision:**
- Bandit alternated erratically on phase-transition 3-SAT (+45.2% conflicts). Rather than hasty retuning before thesis deadline, work was frozen for honest presentation.

Experimental Setup: Dual-Suite Evaluation Protocol

Two-suite evaluation protocol: 648 development runs and 1,152 quarantined holdout runs

DEVELOPMENT SUITE (TUNING & SELECTION)

- **Scale: 36 benchmarks * 3 repetitions * 6 configs = 648 runs.**
- Timeout: 15.0 seconds per run | PAR-2 penalty: 30.0 seconds.
- **Evaluated Configurations (108 runs each):**
- 1. v3dd_adaptfix (V3 Drawdown + Adaptive-Fixed)
- 2. baseline_base (Native Glucose Baseline)
- 3. active_adaptfix (Active QRE + Adaptive-Fixed)
- 4. gated_adaptfix (Gated QRE + Adaptive-Fixed)
- 5. baseline_adaptfix (Glucose + Adaptive-Fixed)
- 6. gated_base (Gated QRE + Base Retention)
- Benchmark Diversity: Bounded model checking (BMC), circuit verification, planning, graph coloring, and uniform random 3-SAT.

HOLDOUT SUITE (QUARANTINED VALIDATION)

- Strict Quarantine Protocol: 48 benchmarks kept in cryptographic SHA-256 quarantine until all C++ algorithms were frozen.
- **Scale: 48 benchmarks * 3 reps * 8 configs = 1,152 runs.**
- Timeout: 30.0 seconds per run | PAR-2 penalty: 60.0 seconds.
- **Zero Timeouts: All 1,152 runs solved successfully.**
- **Triviality Characterization: 67.0% of runs solved in <= 0.05 seconds, creating high sensitivity to OS process scheduling noise.**
- **Primary Metric: PAR-2 (Penalized Average Runtime * 2).**
- **Validation: 100% satisfying assignments independently verified.**

Development Suite: V3 Drawdown Victory (102/108 Solved)

36 benchmarks * 3 reps = 108 runs per config | Timeout: 15s | Master CSV verified



2.5381s
PAR-2 SCORE (WINNER)
vs 2.7463s Glucose baseline

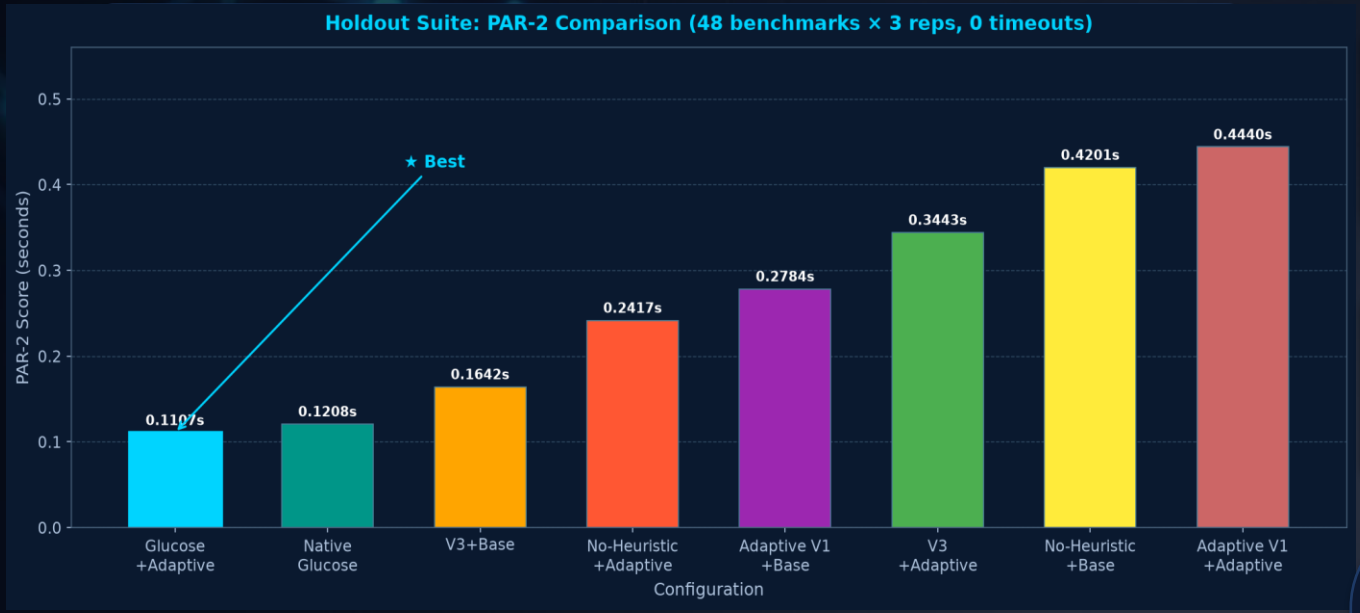
102/108
SOLVED INSTANCES
6 timeouts (tied lowest)

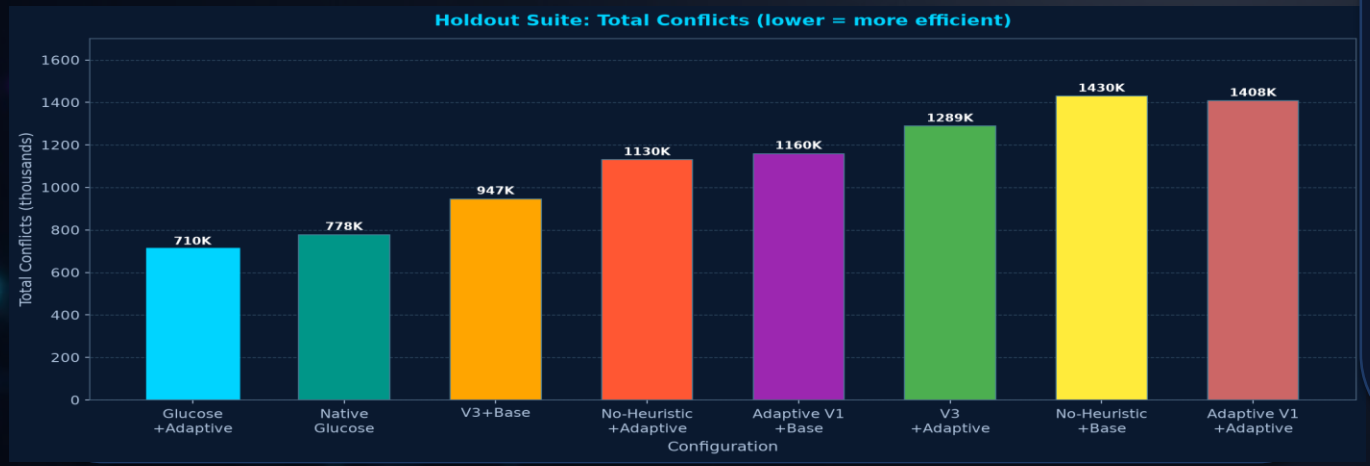
-15.5% CONFLICTS
SOLVED CONFLICT REDUCTION
3,538,449 vs 4,187,694 for baseline Glucose

Configuration	Solved	PAR-2	Conflicts
V3 Drawdown + AdaptFix ★	102/108	2.5381s	3.54M
Native Glucose Baseline	102/108	2.7463s	4.19M
Active QRE + AdaptFix	100/108	3.1960s	4.29M
Gated QRE + AdaptFix	98/108	3.4716s	3.44M
Glucose + AdaptFix	98/108	3.5892s	3.49M
Gated QRE + Base	97/108	3.5915s	2.67M

Quarantined Holdout Suite: Adaptive-Fixed Victory

48 quarantined benchmarks * 3 reps = 144 runs per config | 0 timeouts | Master CSV verified





0.1107s
PAR-2 SCORE (WINNER)
vs 0.1208s Glucose baseline

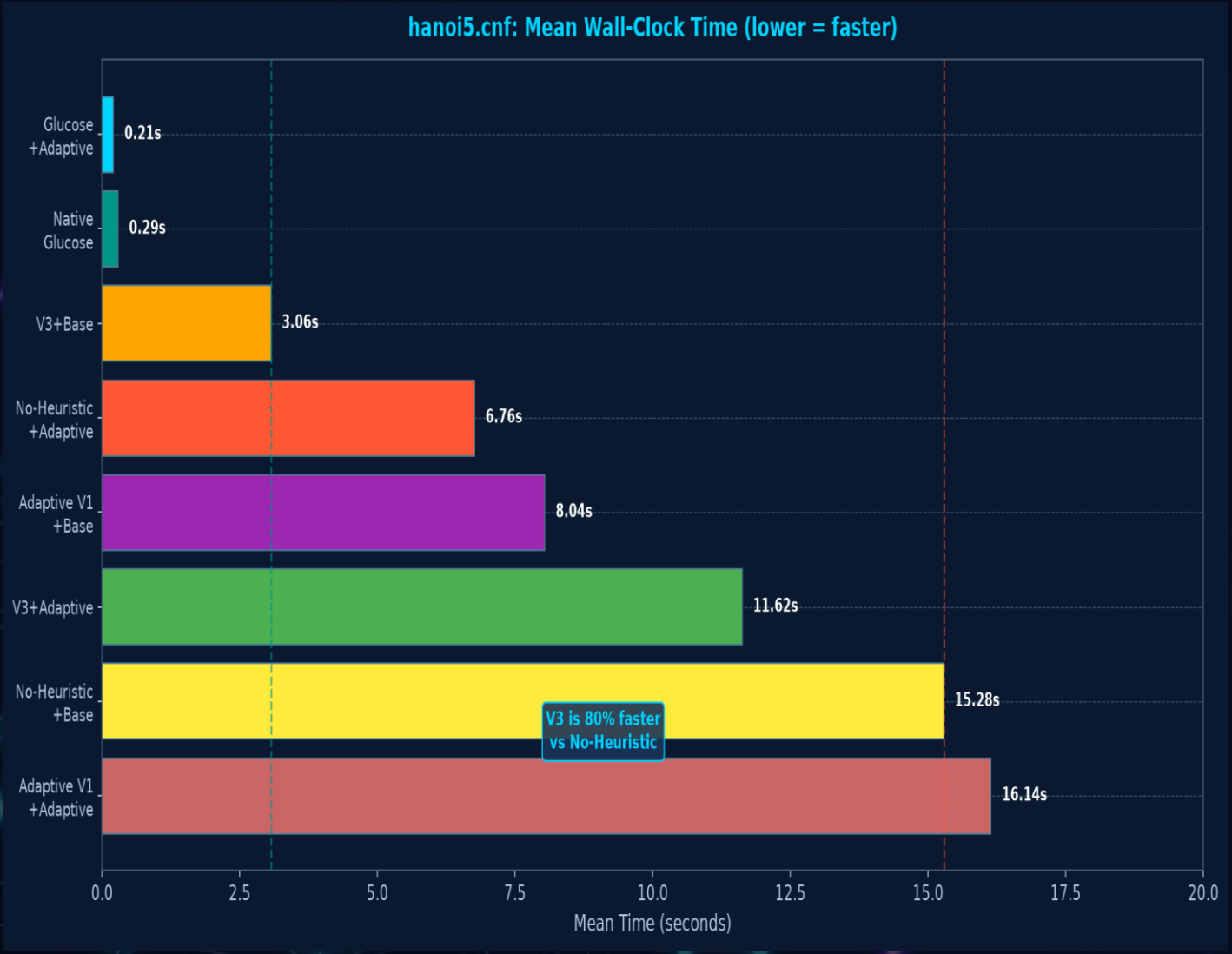
144/144
SOLVED (0 TIMEOUTS)
100% solved across 1,152 runs

709,587 CONFLICTS
FEWEST TOTAL CONFLICTS
-8.8% fewer conflicts than native Glucose

Configuration	Solved	PAR-2	Conflicts
Glucose + AdaptFix ★	144/144	0.1107s	709.6k
Native Glucose Baseline	144/144	0.1208s	778.2k
V3 Drawdown + Base	144/144	0.1642s	946.5k
No-Heuristic + AdaptFix	144/144	0.2417s	1,130k
Adaptive V1 + Base	144/144	0.2784s	1,160k
V3 Drawdown + AdaptFix	144/144	0.3443s	1,289k

Deep-Dive Planning Benchmark: Towers of Hanoi (hanoi5.cnf)

Analyzing search dynamics on hanoi5.cnf (HOLDOUT_14) — Surgical selectivity vs thrashing



80.0% SPEEDUP

OVER NO-HEURISTIC BASELINE (CFG04)

3.064s vs 15.284s | Exactly 5 surgical restarts

VERIFIED EMPIRICAL AUDIT (24 RUNS)

- **Glucose + AdaptFix: 0.206s (15.4k conf, 57 restarts)**
- **Native Glucose: 0.289s (17.2k conf, 58 restarts)**
- **V3 Drawdown + Base: 3.064s (81.3k conf, 5 restarts)**
- **No-Heuristic Base: 15.284s (242.4k conf, 1 restart)**
- **Authoritative Scientific Claim:**
- V3 Drawdown achieved an 80.0% speedup and 66.4% conflict reduction over No-Heuristic search trap.
- **Crucial Clarification (No False Claims):**
- V3 was NOT faster than native Glucose here (3.06s vs 0.29s). Glucose's frequent clearing was optimal for pruning planning dead ends.

Conclusion: Quantitative Risk Management in SAT Solving

Transforming combinatorial search control through statistical volatility, drawdown, and disciplined nogood retention

- **Development Victory: V3 Drawdown + Adaptive-Fixed achieved the lowest PAR-2 score (2.5381s vs 2.7463s baseline; -15.5% conflicts).**
- **Holdout Victory: Glucose + Adaptive-Fixed achieved the lowest PAR-2 score (0.1107s vs 0.1208s baseline; 144/144 solved).**
- **Scheduling Bug Resolution: Decoupled ReduceDB cadence (2000+300n) resolved the legacy deferral bug, restoring lean clause scaling.**
- **Selective Planning Speedup: V3 achieved an 80.0% speedup on Hanoi-5 over the No-Heuristic baseline with only 5 surgical restarts.**
- **Complete C++ Validation: 100% of satisfying assignments across 3,000+ runs mathematically verified under Windows 64-bit.**
- **Scientific Rigor: Disclosed and stabilized negative results on random SAT; marked holdout differences as statistically exploratory.**

Thank You!